Week 2 — The Desk, Then Make It Yours

Mission. Tuesday you build the desk's steps, the data side of Francis & Zhang (2026): from a ticker to the filings, the facts and the ratios, with no model yet, and every number matched to Costco's 10-K after the instructor has run the same steps on the screen. Thursday you add the four roles with your model key: the app, its first five-part report, your audit of it by hand, and the proof on a company your partner picks.

Why this matters

The paper's desk wrote reports that blinded readers rated beside Value Line's. Nobody in the study checked whether the numbers inside were right, and a plausible wrong number is worse than a blank, because nobody checks what looks right. A junior on any desk does two jobs on earnings day: produce the report, and be the reason it can be trusted. You build the first with your agents. The second is yours, by hand, and it is what this course grades.

You arrive with

Your setup checked by a partner in Lab 01, the paper read, and your answer to whether Figure 1 is a workflow or an agent. Your model API key is needed Thursday, not Tuesday.

Outcomes

- Build the desk's seven steps with your AI agents, no model: a ticker in, the filings and the XBRL facts pulled, the ratios computed in code, every number matched to the 10-K.
- Add the four roles on those steps with your model key: a five-part report out.
- Audit a report's numbers against the filings with the report audit, and name a definition trap.
- Run your desk on a company you did not choose, and defend its report to a partner.

The sessions

- **Session 2 —** Lab 02: The desk's steps. The instructor demos the seven steps on the screen; you build them. No model key.
- **Session 3 —** Lab 03: The app. The four roles, the report, the audit. Checkpoint 1 is due by the end of the day.

Read this week

- **Francis & Zhang (2026)**, Section 3.1 again, with Figure 1 and Figure 2's Report A open: the roles, their tools, the JSON hand-off, the five-part template. About 20 minutes. Write one line: which of the paper's tools does your desk get for free from EDGAR?
- The report audit, about 10 minutes.
- **Kim, A. G., Muhn, M., & Nikolaev, V. V. (2024).** *Financial Statement Analysis with Large Language Models.* arXiv:2407.17866. Read the abstract and the first results table, about 20 minutes. Their model called the direction of the next year's earnings change right 60.35% of the time, about 7 percentage points better than analysts. Write one sentence: what would make that edge shrink on companies and years the model never saw?

Teaching 1 — the paper's desk

The paper's input is a ticker. The Strategy Manager turns it into tasks and sends each to a role as JSON: the task, the context the role may use, the output expected. The Research Analyst reads the filings for the business model, competitive position, catalysts and recent developments. The Financial Analyst does the statement analysis: growth, margins, returns, leverage, cash flow, and a comparison with peers. The Investment Advisor synthesizes the two into a thesis and a Buy, Hold or Sell. The Manager checks the five parts of the template are present and consistent, sends a draft back once if not, then compiles the report. In the paper a manager model coordinates the roles; Week 3 returns to that. In your desk this week the Manager is a checklist in code, and that loop is where a desk is made trustworthy or not: the model decides what a sentence says; the harness decides what the model is shown, what it must return, and what happens when it returns something incomplete.

The paper's tools are a filings reader, web search and a calculator. The desk is therefore two things, and this week builds them in that order: the steps, which are code and need no model, and the roles, which are model calls on what the steps produced. EDGAR gives you the filings reader for nothing: the company's filing list, the 10-K and 10-Q documents, the

earnings release attached to the 8-K, and the statements as data through the XBRL company-facts API, every number carrying the tag, the period and the filing it came from. The calculator is your code. Web search is optional this term; nothing graded needs it. Two consequences follow. A desk that works on Costco works on Walmart only if nothing in the harness names Costco: the ticker is an argument, the roles name no company, and the known answer lives in your test. And when a number is wrong, the fix is almost always in a role's instructions or in what the harness checks, not in a smarter model.

The five parts, from the paper's Report A: Overview; Market Positioning and Growth Outlook; Valuation Insights; Future Developments and Strategic Recommendations; Investment Recommendation.

Teaching 2 — the Financial Analyst's numbers

The Financial Analyst is where desks fail, because a model asked for a number produces one whether or not the filing holds it. The rule: every number comes with the filing and the line, or the XBRL tag and period, it came from, and every ratio shows its arithmetic. The check: a person, or a separate role, opens the filing at that line. The report audit fixes eleven core claims for everyone, eight from the latest 10-K and three from the latest earnings release, so that any two desks in the room can be compared.

Definitions are the quiet failure, and the tags carry them. Costco's income statement shows net sales and membership fees on separate lines; total revenue is their sum. Gross margin on net sales, (net sales − merchandise costs) ÷ net sales, and gross margin on total revenue, (total revenue − merchandise costs) ÷ total revenue, are both arithmetic on the same filing, and they differ. Only one is what Costco and its analysts call gross margin. The XBRL facts carry the same trap: a retailer with membership and other income can report total revenues under `Revenues` and net sales under `RevenueFromContractWithCustomerExcludingAssessedTax`, two numbers for the same year. A desk that compares one company's total revenue with another's net sales has compared two definitions. An analyst names the definition beside the number.

Worked example — Costco, from what it filed last

The worked example is a run, not a table. In Session 2 the instructor runs the seven steps on Costco on the screen, one step at a time, and every number comes from what Costco filed last: the 10-K filing list, the 8-K filing list, and the XBRL company facts. Each number is then

found in the filing, at the line the step names. The run is the known answer and the filing is its judge. This page states no figure, because Costco's figures change every time it files; the steps and the checks do not. The check for each step is in Lab 02; the eleven core claims and their definitions are on the report audit, eight from the latest 10-K and three from the latest earnings release, the 8-K's Exhibit 99.1. If you miss the session, your own steps and the filings are the same check.

Valuation needs a price the filings do not carry. Give the desk one, the last close before the release with its date, from the exchange or LSEG Workspace, and the Financial Analyst computes P/E as that price ÷ diluted EPS. From Week 4 your own code supplies the price.

The peers are Walmart, Target and BJ's Wholesale, each read from its own latest 10-K. Their fiscal years end months away from Costco's, so the peer table names the year of each figure, and every margin in it is on one definition. A desk that prints Costco's numbers for Walmart is not reading.

Two traps the steps meet on the way, both real. A filer moves between tags across years, so a reader that takes the first revenue tag it finds can print a year that ended a year ago. The test: every period end in step 4 equals the 10-K's fiscal year end. The fix: among the candidate tags, choose the one whose latest year is newest. And the newest 8-K is often not the earnings release. Companies file 8-Ks for monthly sales, officer changes and shareholder votes, so the earnings 8-K is the newest one that reports Item 2.02.

The fallback

If your desk does not run by a checkpoint's deadline, or your model provider is down, the course still gets a report. Open a chat with any model, paste the income statement, balance sheet and cash flow statement from the latest 10-K and the text of the latest earnings release, and send the four roles' tasks from the Lab 03 request as one message. Save the answer as a Markdown file with the company and the filings named on its first line and "fallback" on its second, and push it. A fallback report counts for Checkpoint 1 and for the live record; it is audited like any other, and the paper says which reports were fallbacks.

What "make it yours" means

Your desk was built from two requests everyone sent. From Week 3 you own its design. The things that are yours: the model and provider each role calls; the words of each role's instructions; the peers the Financial Analyst compares against and the ratios it adds; what the

Manager refuses to compile; the shape of the report within the five parts. The things that are never yours: a company name anywhere in the desk's code (a test may name one), or a number the filings do not state. The test of ownership is simple: your partner names a company that reported this month, you run the same command on its ticker, and the report is about that company, with every number carrying its line.

Before Session 2 — about 60 minutes

1. Finish any Lab 01 row your partner did not initial. Debug with your AI.
2. Watch the Karpathy talk (link on the Week 1 page). Write one line: where is the autonomy slider in your own agent?
3. Read Section 3.1 and Figure 2 of the paper, the report audit and the Kim, Muhn & Nikolaev abstract (above).
4. Open Costco's 10-K (the filing index above) at the income statement. Find net income attributable to Costco and diluted earnings per share. Then open the XBRL company facts (link above) in the browser and search the page for `EarningsPerShareDiluted`: the same number, with its period and filing.

Before Session 3 — about 45 minutes

1. If your steps did not all run in class, get them running with your AI.
2. Run them on WMT with peers of your choice. *Check:* step 4's total revenue and operating income are the figures on the income statement of Walmart's latest 10-K, for its latest fiscal year; the revenue tag printed holds total revenues, not net sales; and the operating margin your calculator prints is the one you get by hand from those two lines.
3. Push your steps and both saved folders' facts to your repository (`git push`, or on github.com **Add file** → **Upload files**, one file at a time; never the `.env` file).
4. Put your model API key in `.env` and confirm `check_env.py` prints its first four characters: Thursday makes the first paid call.

Next: in Week 3 you rebuild the desk as a team whose process is a graph in code, with a verifier no run can skip.